

These estimates are very close to the Fama–French regression estimates. Consistent with Buffett’s avowed eschewal of momentum investing, the UMD loading is close to zero ( $u = -0.04$ ) and statistically insignificant. Note that the adjusted  $R^2$  of this regression is 27%, exactly the same as the Fama–French regression, implying that adding the momentum factor has not improved the fit of the factor regression. Buffett’s alpha has even improved slightly by adding the momentum factor ( $\alpha = 0.68\%$  per month) compared to the Fama–French regression ( $\alpha = 0.65\%$  per month).

For completeness, the mimicking portfolio implied by this Fama–French plus momentum benchmark is:

$$\begin{aligned}
 &(1 - 0.66) = \$0.37 \text{ in T-bills} \\
 &+ \$0.66 \text{ in the market portfolio} \\
 &- \$0.50 \text{ in small caps} + \$0.50 \text{ in large caps} \\
 &+ \$0.36 \text{ in value stocks} - \$0.36 \text{ in growth stocks} \\
 &- \$0.04 \text{ in past winning stocks} + \$0.04 \text{ in past losing stocks}
 \end{aligned}$$

Buffet is also adding:

$$+ 0.68\% \text{ (alpha) per month.}$$

Figure 10.2 shows cumulative excess returns relative to the Fama–French benchmark in the dashed line and the Fama–French plus momentum benchmark in the solid line. Both lie below the CAPM benchmark, which is mainly a consequence of lowering Buffett’s alpha by including the *HML* value-growth factor.

### 3.2 DOING WITHOUT RISK-FREE ASSETS

Benchmark portfolios need not include risk-free assets.

#### *CalPERS*

CalPERS is the largest public pension fund in the United States and had \$246 billion of assets at June 30, 2011.<sup>10</sup> A benchmark for this pension fund might be a passive portfolio of index funds in stocks and bonds—the benchmark that the Canada Pension Plan has adopted through its Reference Portfolio. A stock-bond benchmark can be run extremely cheaply—for close to zero—and is a viable yardstick for judging whether active management is adding value.

A benchmark regression for CalPERS’ returns would be

$$\underbrace{r_{it}}_{\$1} = \alpha + \underbrace{\beta_s r_{st} + \beta_b r_{bt}}_{\$1} + \varepsilon_{it}, \quad (10.12)$$

<sup>10</sup>Data and additional information for this section is from “California Dreamin’: The Mess at CalPERS,” Columbia CaseWorks, #120306 and “Factor Investing: The Reference Portfolio and Canada Pension Plan Investment Board,” Columbia CaseWorks #120302.